UniProt Tools UniProtKB Advanced | List Search Help

P14618 · KPYM_HUMAN

Protein
Pyruvate kinase PKM

Gene
PKM

Status
UniProtKB reviewed (Swiss-Prot)

Organism
Homo sapiens (Human)

Amino acids
531 (go to sequence)

Protein existence
Evidence at protein level

Annotation score
5/5

Tools Download Add to basket Add a publication Entry feedback

Entry Variant viewer Feature viewer Genomic coordinates Publications External links

Function

Catalyzes the final rate-limiting step of glycolysis by mediating the transfer of a phosphoryl group from phosphoenolpyruvate (PEP) to ADP, generating ATP (PubMed:15996096, PubMed:1854723, PubMed:20847263).
The ratio between the highly active tetrameric form and nearly inactive dimeric form determines whether glucose carbons are channeled to biosynthetic processes or used for glycolytic ATP production (PubMed:15996096, PubMed:1854723, PubMed:20847263).
The transition between the 2 forms contributes to the control of glycolysis and is important for tumor cell proliferation and survival (PubMed:15996096, PubMed:1854723, PubMed:20847263). 3 Publications

Isoform M2

Isoform specifically expressed during embryogenesis that has low pyruvate kinase activity by itself and requires allosteric activation by D-fructose 1,6-bisphosphate (FBP) for pyruvate kinase activity (PubMed:18337823, PubMed:20847263).
In addition to its pyruvate kinase activity in the cytoplasm, also acts as a regulator of transcription in the nucleus by acting as a protein kinase (PubMed:18191611, PubMed:21620138, PubMed:22056988, PubMed:22306293, PubMed:22901803, PubMed:24120661).
Translocates into the nucleus in response to various signals, such as EGF receptor activation, and homodimerizes, leading to its conversion into a protein threonine- and tyrosine-protein kinase (PubMed:22056988, PubMed:22306293, PubMed:22901803, PubMed:24120661, PubMed:26787900).
Catalyzes phosphorylation of STAT3 at 'Tyr-705' and histone H3 at 'Thr-11' (H3T11ph), leading to activate

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FOXM1 isoform 5 (FOXM1D) and leads to increased angiogenesis (PubMed:**33314660**).
Its ability to activate transcription plays a role in cancer cells by promoting cell proliferation and promote tumorigenesis (PubMed:**18337823**, PubMed:**22901803**, PubMed:**26787900**).
Promotes the expression of the immune checkpoint protein CD274 in BMAL1-deficient macrophages (By similarity).
May also act as a translation regulator for a subset of mRNAs, independently of its pyruvate kinase activity: associates with subpools of endoplasmic reticulum-associated ribosomes, binds directly to the mRNAs translated at the endoplasmic reticulum and promotes translation of these endoplasmic reticulum-destined mRNAs (By similarity).
Plays a role in caspase independent cell death of tumor cells (PubMed:**17308100**). By Similarity
11 Publications

Isoform M1

Pyruvate kinase isoform expressed in adult tissues, which replaces isoform **M2** after birth (PubMed:**18337823**).
In contrast to isoform **M2**, has high pyruvate kinase activity by itself and does not require allosteric activation by D-fructose 1,6-bisphosphate (FBP) for activity (PubMed:**20847263**). 2 Publications

Miscellaneous

There are 4 isozymes of pyruvate kinase in mammals (L, R, M1, M2) encoded by 2 different genes: PKLR and PKM. The L and R isozymes are generated from the PKLR by differential splicing of RNA; the M1 and M2 forms are produced from the PKM gene by differential splicing. L type is major isozyme in the liver, R is found in red cells, M1 is the main form in muscle, heart and brain, and M2 is found in early fetal tissues as well as in most cancer cells.

Catalytic activity[i]

Rhea:18157

Isoform M2

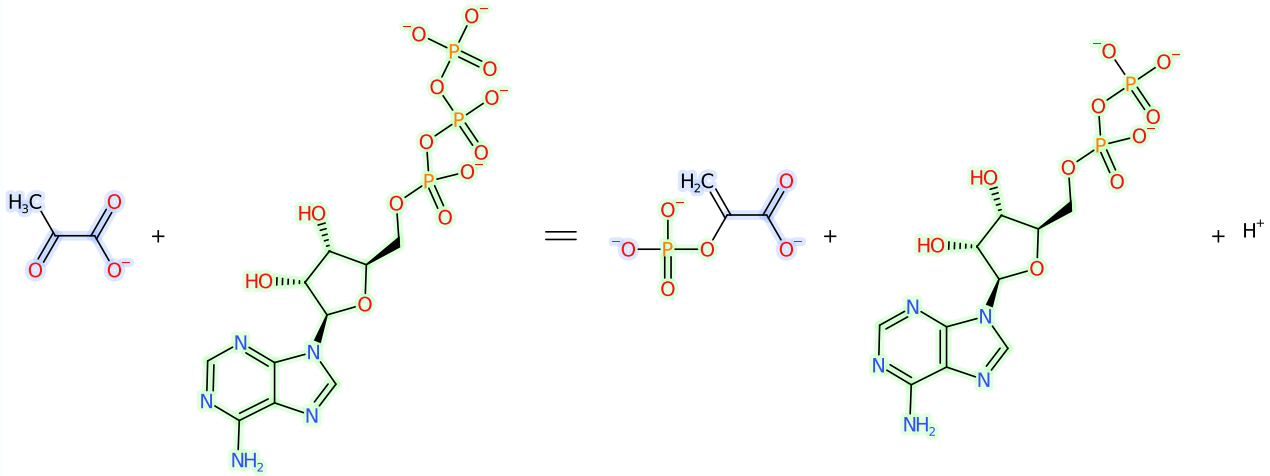
pyruvate + ATP = phosphoenolpyruvate + ADP + H^+ 3 Publications
This reaction proceeds in the backward direction. 3 Publications
EC:2.7.1.40 (**UniProtKB** | **ENZYME** | **Rhea**)

Hide reaction details

Reaction | Atom map

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Color atoms ☑, Number atoms ☐, Confidence = 0.34795



pyruvate » ATP » phosphoenolpyruvate » ADP » H^+ »

Rhea:10596

Isoform M2

L-tyrosyl-[protein] + ATP = O-phospho-L-tyrosyl-[protein] + ADP + H^+ 2 Publications

This reaction proceeds in the forward direction. 2 Publications

EC:2.7.10.2 (**UniProtKB** | **ENZYME** | **Rhea**)

View reaction details

Rhea:46608

Isoform M2

L-threonyl-[protein] + ATP = O-phospho-L-threonyl-[protein] + ADP + H^+ 2 Publications

This reaction proceeds in the forward direction. 2 Publications

EC:2.7.11.1 (**UniProtKB** | **ENZYME** | **Rhea**)

View reaction details

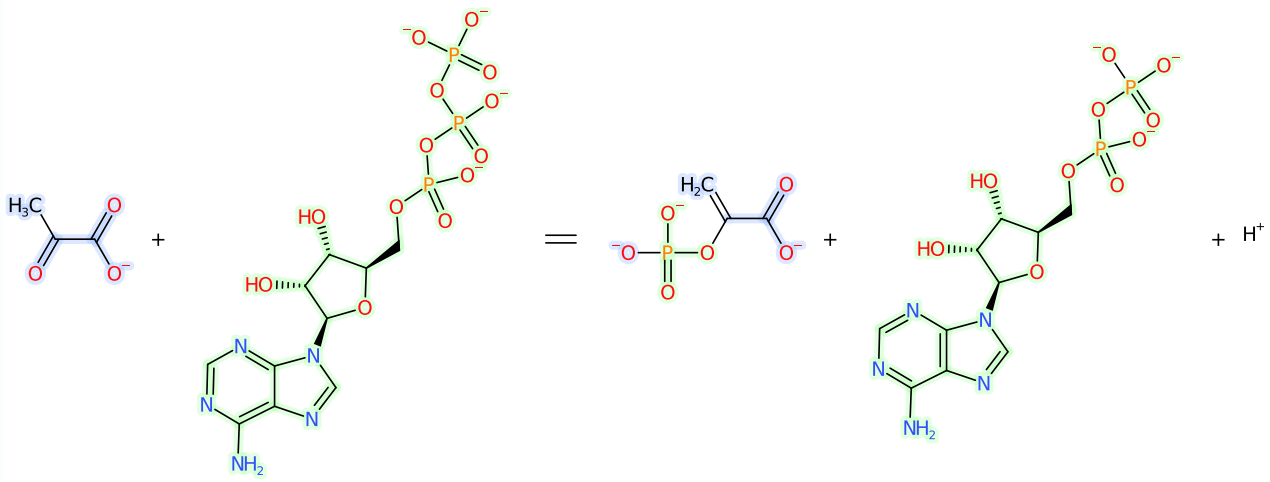
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EC:2.7.1.40 (**UniProtKB** | **ENZYME** | **Rhea**)

Hide reaction details

Reaction **Atom map**

Color atoms ☑, Number atoms ☐, Confidence = 0.34795



pyruvate » ATP » phosphoenolpyruvate » ADP » H^+ »

Cofactor[i]

Protein has several cofactor binding sites:

Mg^{2+} (**UniProtKB** | **Rhea** | **CHEBI:18420**) 1 Publication

K^+ (**UniProtKB** | **Rhea** | **CHEBI:29103**) 1 Publication

Activity regulation[i]

Isoform M2

Isoform **M2** is allosterically activated by D-fructose 1,6-bisphosphate (FBP) (PubMed:**15996096**,

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inhibition of PKM enzymatic activity, this diverts glucose metabolites from energy production to anabolic processes when cells are stimulated by certain growth factors (PubMed:**18337815**).
Glycolytic flux are highly dependent on de novo biosynthesis of serine and glycine, and serine is a natural ligand and allosteric activator of isoform **M2** (PubMed:**23064226**).
Acetylation at Lys-433 promotes its translocation into the nucleus and homodimerization, promoting the protein kinase activity (PubMed:**24120661**). 7 Publications

Isoform M1

Has high pyruvate kinase activity by itself and does not require allosteric activation by D-fructose 1,6-bisphosphate (FBP) for activity. 1 Publication

Kinetics[i]

KM	SUBSTRATE	PH	TEMPERATURE[C]	NOTES	EVIDENCE
2.7 mM	phosphoenolpyruvate	8.0	32		2 Publications
0.17 mM	phosphoenolpyruvate	8.0	32	in the presence of 2 mM D-fructose 1,6-bisphosphate (FBP),	2 Publications
0.34 mM	ADP	8.0	32		2 Publications
0.24 mM	ADP	8.0	32	in the presence of 2 mM FBP,	2 Publications
0.13 mM	phosphoenolpyruvate		25	in the presence of 2 mM FBP	2 Publications
0.63 mM	ADP		25	in the presence of 2 mM FBP	2 Publications

pH Dependence[i]

canonical

Optimum pH for T3 binding is 6.0-6.5. Increase in pH causes T3 binding to drop, does not bind T3 above pH 9.0 or below pH 5.0. 2 Publications

Pathway[i]

Carbohydrate degradation; glycolysis; pyruvate from D-glyceraldehyde 3-phosphate: step 5/5. 2 Publications

Features

Showing features for binding site[i], site[i].

± ▾ TYPE All ID POSITION(S) DESCRIPTION

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	TYPE	ID	POSITION(S)	DESCRIPTION	
+	Binding site		75	K^+ (UniProtKB | ChEBI) 1 Publication Combined Sources	
+	Binding site		75-78	ATP (UniProtKB | ChEBI) 1 Publication Combined Sources	Tools Add

Expand table

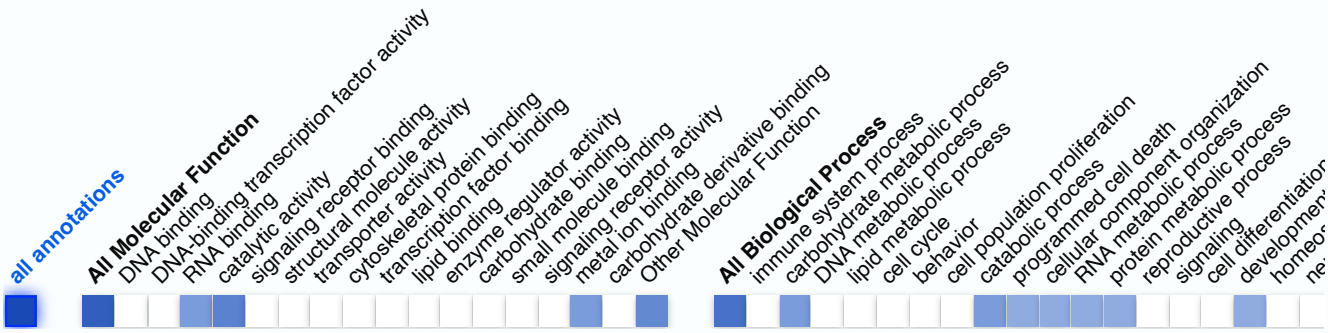
Gene Ontology[i]

GO annotations | GO-CAM models New

Gene Ontology (GO) annotations organized by slimming set.

Slimming set:

Animals and fungi (Opisthokonta)



Cell color indicative of number of GO terms

ASPECT	TERM
Molecular Function	cadherin binding Source:BHF-UCL 1 Publication
Molecular Function	histone H3T11 kinase activity Source:UniProtKB
Molecular Function	magnesium ion binding Source:InterPro
Molecular Function	MHC class II protein complex binding Source:UniProtKB 1 Publication
Molecular	

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ASPECT	TERM

Expand table

Access the complete set of GO annotations on QuickGO

PAN-GO
P14618 4 GO annotations based on evolutionary models

Keywords[i]

Molecular function
#Allosteric enzyme
#Kinase
#Transferase

Biological process
#Glycolysis
#Translation regulation

Ligand
#ATP-binding
#Magnesium
#Metal-binding
#Nucleotide-binding
#Potassium
#Pyruvate

Enzyme and pathway databases

BRENDA
2.7.1.40 2681

BioCyc
MetaCyc:HS00906-MONOMER

NDEx
IQUERY-CP-PKM 9 NDEx IQuery Curated Pathways

PathwayCommons
P14618

Reactome
R-HSA-6798695 Neutrophil degranulation
R-HSA-70171 Glycolysis
R-HSA-70268 Pyruvate metabolism
R-HSA-9764725 Negative Regulation of CDH1 Gene Transcription
R-HSA-9861718 Regulation of pyruvate metabolism [**P14618-1**]

SABIO-RK
P14618

SIGNOR
P14618

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UniPathway
UPA00109UER00188

ENZYME
Search...

Protein family/group databases

MoonDB
P14618 Curated

MoonProt
P14618

TCDB
8.A.261.1.1 the pkm2 pyruvate kinase regulator ({pkm2) family

Names & Taxonomy

Protein names

Recommended name
Pyruvate kinase PKM

EC number
EC:2.7.1.40 (UniProtKB | ENZYME | Rhea) 3 Publications

Alternative names
Cytosolic thyroid hormone-binding protein 1 Publication (CTHBP 1 Publication)
Opa-interacting protein 3 1 Publication (OIP-3 1 Publication)
Pyruvate kinase 2/3
Pyruvate kinase muscle isozyme
Threonine-protein kinase PKM2 Curated (EC:2.7.11.1 2 Publications) . EC:2.7.11.1 (UniProtKB | ENZYME | Rhea) 2 Publications
More alternative names

Gene names

Name
PKM

Synonyms
OIP3 1 Publication, PK2, PK3, PKM2

Organism names

Taxonomic identifier
9606 (NCBI)

Organism
Homo sapiens (Human)

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Eukaryota > Metazoa > Chordata > Craniata > Vertebrata > Euteleostomi > Mammalia > Eutheria > Euarchontoglires > Primates > Haplorrhini > Catarrhini > Hominidae > Homo

Accessions[i]

Primary accession

P14618

Secondary accessions

A6NFK3
B2R5N8
B3KRY0
B4DFX8
B4DUU6

More accessions

Proteomes[i]

Identifier

UP000005640

Component[i]

Chromosome 15

Organism-specific databases

AGR
HGNC:9021

HGNC
HGNC:9021 PKM

MIM
179050 gene

VEuPathDB
HostDB:ENSG00000067225

Subcellular Location[i]

UniProt Annotation | GO Annotation

Isoform M2

Cytoplasm 4 Publications

Nucleus 8 Publications

Note: Translocates to the nucleus in response to various signals, such as EGF receptor activation or apoptotic stimuli (PubMed:17308100, PubMed:22056988, PubMed:24120661).

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Isoform M1

Cytoplasm Curated

Keywords[i]

Cellular component

#Cytoplasm

#Nucleus

Miscellaneous

CD-CODE

91857CE7 Nucleolus

FB4E32DD Presynaptic clusters and postsynaptic densities

Disease & Variants[i]

Features

Showing features for natural variant[i], mutagenesis[i].

±	TYPE	ID	POSITION(S)	DESCRIPTION
+	Natural variant	VAR_033067	204	in dbSNP:rs17853396 1 Publication
+	Mutagenesis		367	Abolishes both pyruvate kinase and protein kinase activities. 2 Publications
+	Mutagenesis		399	Impaired homotetramerization, leading to homodimerization and subsequent activation of the protein kinase activity. 1 Publication
+	Mutagenesis		403	Significant reduction in hydroxylation and in PKM-mediated transcriptional activity of HIF1A; when associated with A-408. 1 Publication
				Significant reduction in hydroxylation

Expand table

Organism-specific databases

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MalaCards
PKM

OpenTargets
ENSG00000067225

Miscellaneous

Pharos
P14618 Tchem

Chemistry

ChEMBL
CHEMBL1075189

DrugCentral
P14618

DrugBank
DB07697 1-(2,3-dihydro-1,4-benzodioxin-6-ylsulfonyl)-4-[(4-methoxyphenyl)sulfonyl]piperazine
DB07692 1-[(2,6-difluorophenyl)sulfonyl]-4-(2,3-dihydro-1,4-benzodioxin-6-ylsulfonyl)piperazine
DB02726 2-Phosphoglycolic Acid
DB07628 6-(2-fluorobenzyl)-2,4-dimethyl-4,6-dihydro-5H-thieno[2',3':4,5]pyrrolo[2,3-d]pyridazin-5-one
DB00787 Acyclovir
More DrugBank links

GuidetoPHARMACOLOGY
3006

Genetic variation databases

BioMuta
PKM

ClinGen
HGNC:9021

DMDM
20178296

GenCC
HGNC:9021

Expression[i]

Tissue specificity[i]

Isoform M2

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Expressed in adult tissues (PubMed:**18337823**).
Not expressed in tumor cells (PubMed:**18337823**). 1 Publication

Developmental stage[i]

Isoform M2

Specifically expressed during embryonic development. 1 Publication

Isoform M1

Specifically expressed in adult tissues. 1 Publication

Gene expression databases

Bgee

ENSG00000067225 Expressed in right hemisphere of cerebellum and 208 other cell types or tissues

ExpressionAtlas

P14618 baseline and differential

Organism-specific databases

HPA

ENSG00000067225 Tissue enhanced (skeletal muscle, tongue)

Interaction[i]

Subunit[i]

Isoform M2

Monomer and homotetramer; exists as a monomer in the absence of D-fructose 1,6-bisphosphate (FBP), and reversibly associates to form a homotetramer in the presence of FBP (PubMed:**15996096**, PubMed:**18298799**, PubMed:**18337815**, PubMed:**1854723**, PubMed:**23064226**, PubMed:**2813362**).
The monomeric form binds 3,3',5-triiodo-L-thyronine (T3) (PubMed:**15996096**).
Tetramer formation induces pyruvate kinase activity (PubMed:**15996096**, PubMed:**18298799**, PubMed:**18337815**, PubMed:**1854723**, PubMed:**23064226**, PubMed:**2813362**).
The tetrameric form has high affinity for the substrate and is associated within the glycolytic enzyme complex (PubMed:**15996096**, PubMed:**18298799**, PubMed:**18337815**, PubMed:**1854723**, PubMed:**23064226**, PubMed:**2813362**).
FBP stimulates the formation of tetramers from dimers (PubMed:**15996096**, PubMed:**18298799**, PubMed:**18337815**, PubMed:**1854723**, PubMed:**23064226**, PubMed:**2813362**).
Homodimer; exists in a dimeric form in tumor cells and the dimeric form has less affinity for the phosphoenolpyruvate substrate (PubMed:**22306293**, PubMed:**24120661**).
The homodimer converts into a protein kinase (PubMed:**22306293**, PubMed:**24120661**).
Interacts with HERC1, POU5F1 and PML (PubMed:**12650930**, PubMed:**18191611**).

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activity under hypoxia (PubMed:**21620138**).
Interacts with TRIM35; this interaction prevents FGFR1-dependent tyrosine phosphorylation (PubMed:**25263439**).
Interacts with JMJD8 (PubMed:**27199445**).
Interacts with TRAF4 (PubMed:**32268273**).
Interacts with (phosphorylated) CTNNB1; leading to activate transcription (PubMed:**22056988**).
Interacts with TSC22D2; the interaction results in reduced nuclear levels of PKM isoform **M2**, leading to repression of cyclin CCND1 transcription and reduced cell growth (PubMed:**27573352**).
Interacts with FOXM1 isoform 5 (FOXM1D); FOXM1D binds to the PKM2 homotetramer to form a heterooctamer, consisting of the PKM2 homotetramer with 4 copies of FOXM1, which reduces PKM2 activity, leading to promotion of glycolysis (PubMed:**33314660**).
Interaction with FOXM1D also promotes nuclear translocation of PKM2 along with the NFKB1/p50 and RELA/p65 subunits of the NF-kappa-B transcription factor complex, leading to increased VEGFA transcription and angiogenesis (PubMed:**33314660**). 18 Publications

(Microbial infection) Binding to certain oncoproteins such as HPV-16 E7 oncoprotein promotes homodimerization. 1 Publication

Binary interactions[i]

TYPE	ENTRY 1	ENTRY 2	NUMBER OF EXPERIMENTS	INTACT
BINARY	P14618	ARRB1 P49407	3	EBI-353408, EBI-743313
BINARY	P14618	ARRB2 P32121	4	EBI-353408, EBI-714559
BINARY	P14618	BOD1 Q96IK1-2	3	EBI-353408, EBI-18924329
BINARY	P14618	CTNNB1 P35222	4	EBI-353408, EBI-491549
BINARY	P14618	DAPK1 P53355	3	EBI-353408, EBI-358616
BINARY	P14618	FGFR3	3	EBI-353408, EBI-

Expand table

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CPX-3057 PKM2 pyruvate kinase complex (dimer)

View interactors in UniProtKB
View CPX-3057 in Complex Portal

Protein-protein interaction databases

BioGRID
111332 548 interactors

CORUM
P14618

ComplexPortal
CPX-3057 PKM2 pyruvate kinase complex (dimer) [P14618-1]
CPX-3058 PKM2 pyruvate kinase complex (tetramer) [P14618-1]
CPX-3093 PKM1 pyruvate kinase complex [P14618-2]

DIP
DIP-31273N

FunCoup
P14618 1373 interactors

IntAct
P14618 337 interactors

MINT
P14618

STRING
9606.ENSP00000320171

Chemistry

BindingDB
P14618

Miscellaneous

RNAct
P14618 protein

Structure[i]

3D structure databases

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EMDB
EMD-10575
EMD-10576
EMD-10577
EMD-10584

MDRepo
P14618

MDposit
P14618

SMR
P14618

ModBase
Search...

PDBe-KB
Search...

Miscellaneous

EvolutionaryTrace
P14618

Family & Domains[i]

Features

Showing features for region[i].

±	TYPE	ID	POSITION(S)	DESCRIPTION	
+	Region		307-531	Interaction with POU5F1 1 Publication	Tools Add
+	Region		389-433	Intersubunit contact	Tools Add

Sequence similarities[i]

Belongs to the **pyruvate kinase family**. Curated

Family and domain databases

- View the Phylogenomic databases for this entry within the **Homologs section**.
- View all family and domain features for this entry's canonical sequence in the **UniParc Feature Viewer**.

CDD
cd00288 Pyruvate_Kinase 1 hit

FunFam
3.20.20.60:FF:000025 Pyruvate kinase 1 hit

Gene3D
3.20.20.60 Phosphoenolpyruvate-binding domains 1 hit
2.40.33.10 PK beta-barrel domain-like 1 hit
3.40.1380.20 Pyruvate kinase, C-terminal domain

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InterPro
View protein in InterPro
IPR001697 Pyr_Knase
IPR015813 Pyrv/PenolPyrv_kinase-like_dom
IPR040442 Pyrv_kinase-like_dom_sf
IPR011037 Pyrv_Knase-like_insert_dom_sf
More InterPro links

NCBIfam
NF004491 PRK05826.1 1 hit
NF004978 PRK06354.1 1 hit
TIGR01064 pyruv_kin 1 hit

PANTHER
PTHR11817 PYRUVATE KINASE 1 hit

PRINTS
PR01050 PYRUVTKNASE

PROSITE
View protein in PROSITE
PS00110 PYRUVATE_KINASE 1 hit

Pfam
View protein in Pfam
PF00224 PK 1 hit
PF02887 PK_C 1 hit

SUPFAM
SSF51621 Phosphoenolpyruvate/pyruvate domain 1 hit
SSF50800 PK beta-barrel domain-like 1 hit
SSF52935 PK C-terminal domain-like 1 hit

MobiDB
Search…

Sequence & Isoforms[i]

Align isoforms (3) **Add isoforms**

Sequence status[i]
Complete

This entry describes **3** isoforms[i] produced by **Alternative splicing**.

P14618-1

This isoform has been chosen as the **canonical** sequence. All positional information in this entry refers to it. This is also the sequence that appears in the downloadable versions of the entry.

Name
M2
Synonyms
M2-PK, PKM2

See also
sequence in **UniParc** or sequence clusters in **UniRef**

Length
531

Mass (Da)
57,937

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B9F341DB7EDC1C70C1439F8176EA2889

MSKPHSEAGTAFIQTQQLHAAMADTFLEHMCRLDIDSPPITARNTGIICTIGPASRSVETLKEMIKSGMNVARLN
FSHGTHEYHAETIKNVRTATESFASDPILYRPVAVALDTKGPEIRTGLIKGSGTAEVELKKGATLKITLDNAYMEK
CDENILWLDYKNICKVVEVGSKIYVDDGLISLQVKQKGADFLVTEVENGGSLGSKKGVNLPGAAVDLPAVSEKD
IQDLKFGVEQDVDMVFASFIRKASDVHEVRKVLGEKGKNIKIISKIENHEGVRRFDEILEASDGIMVARGDLGIEI
PAEKVFLAQKMMIGRCNRAGKPVICATQMLESMIKKPRPTRAEGSDVANAVLDGADCIMLSGETAKGDYPLEA
VRMQHLIAREAEAAIYHLQLFEELRRLAPITSDPTEATAVGAVEASFKCCSGAIIVLTKSGRSAHQVARYRPRAPI
IAVTRNPQTARQAHLYRGIFPVLCKDPVQEAWAEDVDLRVNFAMNVGKARGFFKKGDVVIVLTGWRPGSGFT
NTMRVVPVP

P14618-2

Name
M1
Synonyms
M1-PK, PKM1
See also
sequence in **UniParc** or sequence clusters in **UniRef**

Differences from canonical
389-433:
IYHLQLFEELRRLAPITSDPTEATAVGAVEASFKCCSG AIIVLTK → MFHRKLFEELVRASSHSTDLMEAMAMGSVEASYKCLAA ALIVLTE 1 Publication

P14618-3

Name
3
See also
sequence in **UniParc** or sequence clusters in **UniRef**

Differences from canonical
1-82:
MSKPHSEAGTAFIQTQQLHAAMADTFLEHMCRLDIDSP PITARNTGIICTIGPASRSVETLKEMIKSGMNVARLNF SHGTHE → MSPEAQPQRTKGPQQPCRSPIVKPGLPSFRPSSCTQPW LTHSWSTCAAWTLIHHPSQPGTLASSVPL
1 Publication

Computationally mapped potential isoform sequences[i]

There are 34 potential isoforms mapped to this entry

Align Add **View all**

Entry	Entry name	Gene name	Length
A0ACI8QYS9	A0ACI8QYS9_HUMAN	PKM	529
A0A0A0MRZ0	A0A0A0MRZ0_HUMAN	PKM	457
A0A8V8TNX9	A0A8V8TNX9_HUMAN	PKM	605
A0A8V8TMM5	A0A8V8TMM5_HUMAN	PKM	344

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Entry	Entry name	Gene name	Length
A0ACI8R4K5	A0ACI8R4K5_HUMAN	PKM	492
A0A804F6T5	A0A804F6T5_HUMAN	PKM	566
A0A804F729	A0A804F729_HUMAN	PKM	605
A0ACI8TPY3	A0ACI8TPY3_HUMAN	PKM	275
A0ACI8TPP9	A0ACI8TPP9_HUMAN	PKM	528
A0ACI8TPV6	A0ACI8TPV6_HUMAN	PKM	468
A0ACI8TPR7	A0ACI8TPR7_HUMAN	PKM	531
A0ACI8TPN7	A0ACI8TPN7_HUMAN	PKM	527
A0ACI8TPQ8	A0ACI8TPQ8_HUMAN	PKM	525
A0ACI8TR48	A0ACI8TR48_HUMAN	PKM	525
A0ACI8TQL3	A0ACI8TQL3_HUMAN	PKM	266
A0ACI8TQG1	A0ACI8TQG1_HUMAN	PKM	348
A0ACI8TRS5	A0ACI8TRS5_HUMAN	PKM	222
A0ACI8TRQ0	A0ACI8TRQ0_HUMAN	PKM	582
A0ACI8TRS3	A0ACI8TRS3_HUMAN	PKM	506
A0ACI8TRF9	A0ACI8TRF9_HUMAN	PKM	502
H3BUW1	H3BUW1_HUMAN	PKM	162
H3BU13	H3BU13_HUMAN	PKM	82
H3BTN5	H3BTN5_HUMAN	PKM	485
H3BTJ2	H3BTJ2_HUMAN	PKM	168
H3BT25	H3BT25_HUMAN	PKM	151
H3BQZ3	H3BQZ3_HUMAN	PKM	69
H3BQ34	H3BQ34_HUMAN	PKM	281
H3BN34	H3BN34_HUMAN	PKM	51

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Entry	Entry name	Gene name	Length
A0ACI8VQD3	A0ACI8VQD3_HUMAN	PKM	422
A0ACI8VQH4	A0ACI8VQH4_HUMAN	PKM	506
A0ACI8VQI2	A0ACI8VQI2_HUMAN	PKM	593

Sequence caution[i]

The sequence BAG57589.1 differs from that shown. Reason: Erroneous initiation Extended N-terminus.
Curated

Features

Showing features for alternative sequence[i], sequence conflict[i].

±	TYPE	ID	POSITION(S)	DESCRIPTION	
+	Alternative sequence	VSP_043370	1-82	in isoform 3 1 Publication	Tools Add
+	Sequence conflict		7	in Ref. 10; AAH12811 Curated	
+	Sequence conflict		54	in Ref. 5; BAG52542 Curated	
+	Sequence conflict		103	in Ref. 2; AAA36672 Curated	

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Keywords[i]

Coding sequence diversity

#Alternative splicing

Technical term

#3D-structure
#Direct protein sequencing
#Proteomics identification
#Reference proteome

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CCDS
CCDS32284.1 [P14618-1]
CCDS32285.1 [P14618-2]

PIR
S30038 S30038
S64635 S64635

RefSeq
NP_001193725.1 NM_001206796.3
NP_001193726.1 NM_001206797.3
NP_001193727.1 NM_001206798.3 [P14618-3]
NP_001193728.1 NM_001206799.2
NP_001303247.1 NM_001316318.2
More RefSeq links

NUCLEOTIDE SEQUENCE	PROTEIN SEQUENCE	MOLECULE TYPE	STATUS
M23725 EMBL · GenBank · DDBJ	AAA36449.1 EMBL · GenBank · DDBJ	mRNA	
M26252 EMBL · GenBank · DDBJ	AAA36672.1 EMBL · GenBank · DDBJ	mRNA	
X56494 EMBL · GenBank · DDBJ	CAA39849.1 EMBL · GenBank · DDBJ	Genomic DNA	
AK092369 EMBL · GenBank · DDBJ	BAG52542.1 EMBL · GenBank · DDBJ	mRNA	
AK222927	BAD96647.1		

Expand table

Genome annotation databases

Ensembl
ENST00000319622.10 ENSP00000320171.6
ENSG00000067225.22 [P14618-2]
ENST00000335181.10 ENSP00000334983.5
ENSG00000067225.22 [P14618-1]
ENST00000568459.5 ENSP00000456970.1
ENSG00000067225.22 [P14618-2]
ENST00000891959.1 ENSP00000562018.1
ENSG00000067225.22 [P14618-2]
ENST00000891960.1 ENSP00000562019.1
ENSG00000067225.22 [P14618-1]
More Ensembl links

GeneID
5315

KEGG
hsa:5315

MANE-Select
ENST00000335181.10 ENSP00000334983.5
NM_002654.6 NP_002645.3

UCSC
uc002atw.2 human [P14618-1]

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Similar Proteins[i]

UniRef clusters[i]



Homologs[i]

Orthologs & paralogs[i]



Phylogenomic databases

GeneTree
ENSGT00390000008859

HOGENOM
CLU_015439_0_1_1

InParanoid
P14618

OMA
RVHHIGE

OrthoDB
108365at2759

PAN-GO
P14618 4 GO annotations based on evolutionary models

PhylomeDB
P14618

eggNOG
KOG2323 Eukaryota

Organism-specific databases

AGR
HGNC:9021

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Species (Proteomes)	Taxonomy	Align	About & Help
Protein clusters (UniRef)	Keywords	Retrieve/ID mapping	UniProtKB manual
Sequence archive (UniParc)	Subcellular locations	Peptide search	Technical corner
	Cross-referenced databases	Tool results	Expert biocuration
	Diseases		Statistics

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